Functional Equations with Inequalities

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1 Examples

IMO Shortlist 2023 A2

Example 1: Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that

$$f(x+y)f(x-y) \geqslant f(x)^2 - f(y)^2$$

for every $x, y \in \mathbb{R}$. Assume that the inequality is strict for some $x_0, y_0 \in \mathbb{R}$.
Prove that either $f(x) \geqslant 0$ for every $x \in \mathbb{R}$ or $f(x) \leqslant 0$ for every $x \in \mathbb{R}$.

Proof:

$$P\left(\frac{a+b}{2}, \frac{a-b}{2}\right) \implies f(a)f(b) \geq f\left(\frac{a+b}{2}\right)^2 - f\left(\frac{a-b}{2}\right)^2$$

Let this expression be $Q(a, b)$.

$$Q(a, -b) \implies f(a)f(-b) \geq -\left(f\left(\frac{a+b}{2}\right)^2 - f\left(\frac{a-b}{2}\right)^2\right)$$

Adding gives $f(a)(f(b) + f(-b)) \geq 0$

Now, $Q(x_0, y_0) \implies f(x_0)(f(y_0) + f(-y_0)) > 0$.

If $f(x_0) > 0$, $f(y_0) + f(-y_0) > 0$ so $Q(a, y_0) \implies f(a) \geq 0$

If $f(x_0) < 0$, $f(y_0) + f(-y_0) < 0$ so $Q(a, y_0) \implies f(a) \leq 0$.

AoPS solution: me

IMO Shortlist 2023 A4

Example 2: Determine all functions $f : \mathbb{R}_{>0} \to \mathbb{R}_{>0}$ such that

$$x\big(f(x) + f(y)\big) \geqslant \big(f(f(x)) + y\big)f(y)$$

for every $x, y \in \mathbb{R}_{>0}$.

Proof: Let $P(x, y)$ denote the assertion.

$P(x, x) \implies x \geq f(f(x))$

Assume for the sake of contradiction $f(f(x)) < x$ for some $x \in \mathbb{R}_+$. Let $x - f(f(x)) = z > 0$.

$$P(f^n(x), f^{n-1}(x)) \implies f^{n+1}(x) - f^{n-1}(x) \geq f^{n+2}(x) - f^n(x)$$

So,

$$x - f(f(x)) \geq f(x) - f(f(f(x))) \geq \cdots \geq f^{n+1}(x) - f^{n-1}(x) \geq f^{n+2}(x) - f^n(x)$$
$$\implies x - f(f(x)) + f(f(x)) - f(f(f(f(x)))) + \cdots + f^{2k-2}(x) - f^{2k}(x) = x - f^{2k}(x) \geq kz$$

$x > x - f^{2k}(x) \geq kz$. However, since $z > 0$, this is a contradiction for large enough k.

So, $f(f(x)) = x \implies xf(x) \geq yf(y)$

Switching x and y yields $xf(x) = yf(y)$, so $xf(x) = c$ and $f(x) = \frac{c}{x}$, which works $\forall c > 0$.

AoPS solution: me

2 Practise Problems

Here are some practise problems, in increasing order of difficulty.

1. **IMO Shortlist 2013 A3**

   Let $f : \mathbb{Q}_{>0} \to \mathbb{R}$ satisfy

   $$f(x)f(y) \geq f(xy), \qquad f(x+y) \geq f(x) + f(y)$$

   for all positive rational x, y. If $f(a) = a$ for some rational $a > 1$, prove that $f(x) = x$ for all positive rational x.

2. **IMO Shortlist 2022 A3**

   Find all functions $f : \mathbb{R}_{>0} \to \mathbb{R}_{>0}$ such that, for each $x > 0$, there is exactly one $y > 0$ satisfying

   $$xf(y) + yf(x) \leq 2.$$

3. **IMO Shortlist 2011 A6**

   Let $f : \mathbb{R} \to \mathbb{R}$ satisfy

   $$f(x+y) \leq yf(x) + f(f(x))$$

   for all real x, y. Prove that $f(x) = 0$ for every $x \leq 0$.

4. **IMO Shortlist 2017 A8**

   Assume that $f : \mathbb{R} \to \mathbb{R}$ has the following property: whenever $x, y \in \mathbb{R}$ satisfy

   $$(f(x) + y)(f(y) + x) > 0,$$

   one has $f(x) + y = f(y) + x$. Prove that

   $$f(x) + y \leq f(y) + x \qquad \text{whenever } x > y.$$